COMP 3613 Assignment 1

Individual report

Student Awards Testing Assignment

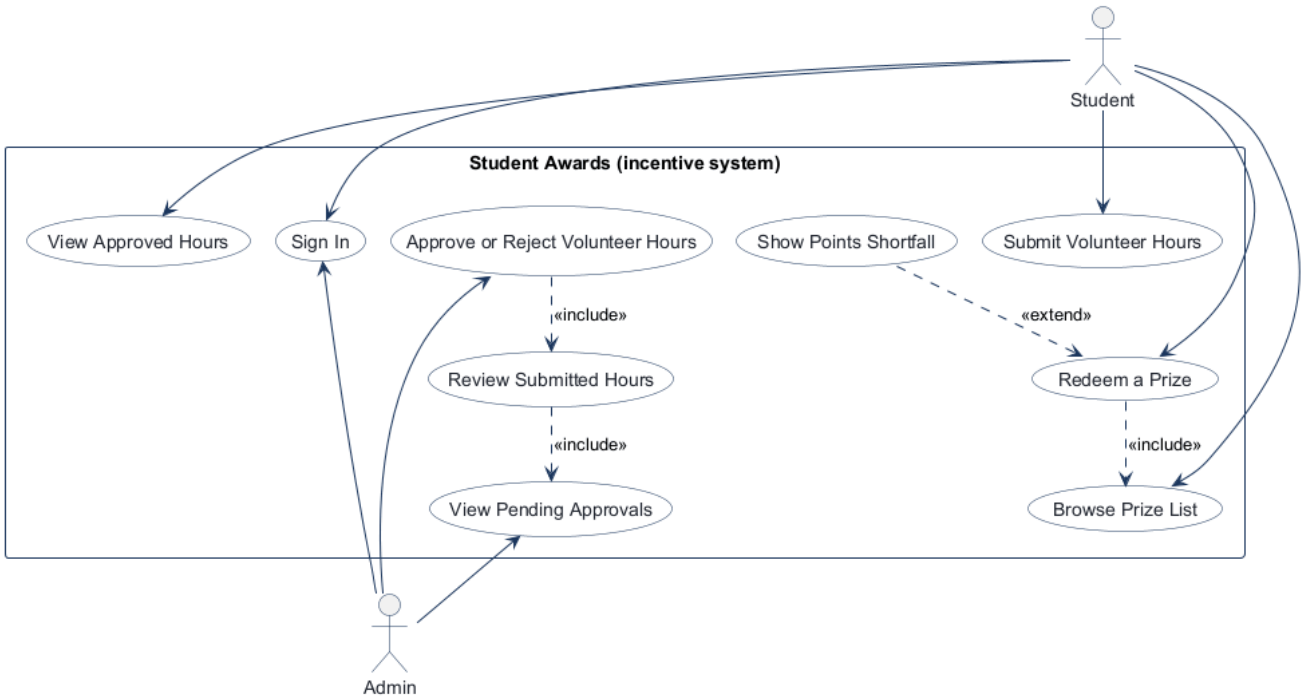
Project testing deliverables: plans, implementation summary, user acceptance testing, automated test reports, and Postman API evidence.

Student	Details
Name	Aliyah Ruan
Student ID	816044540

Project Description

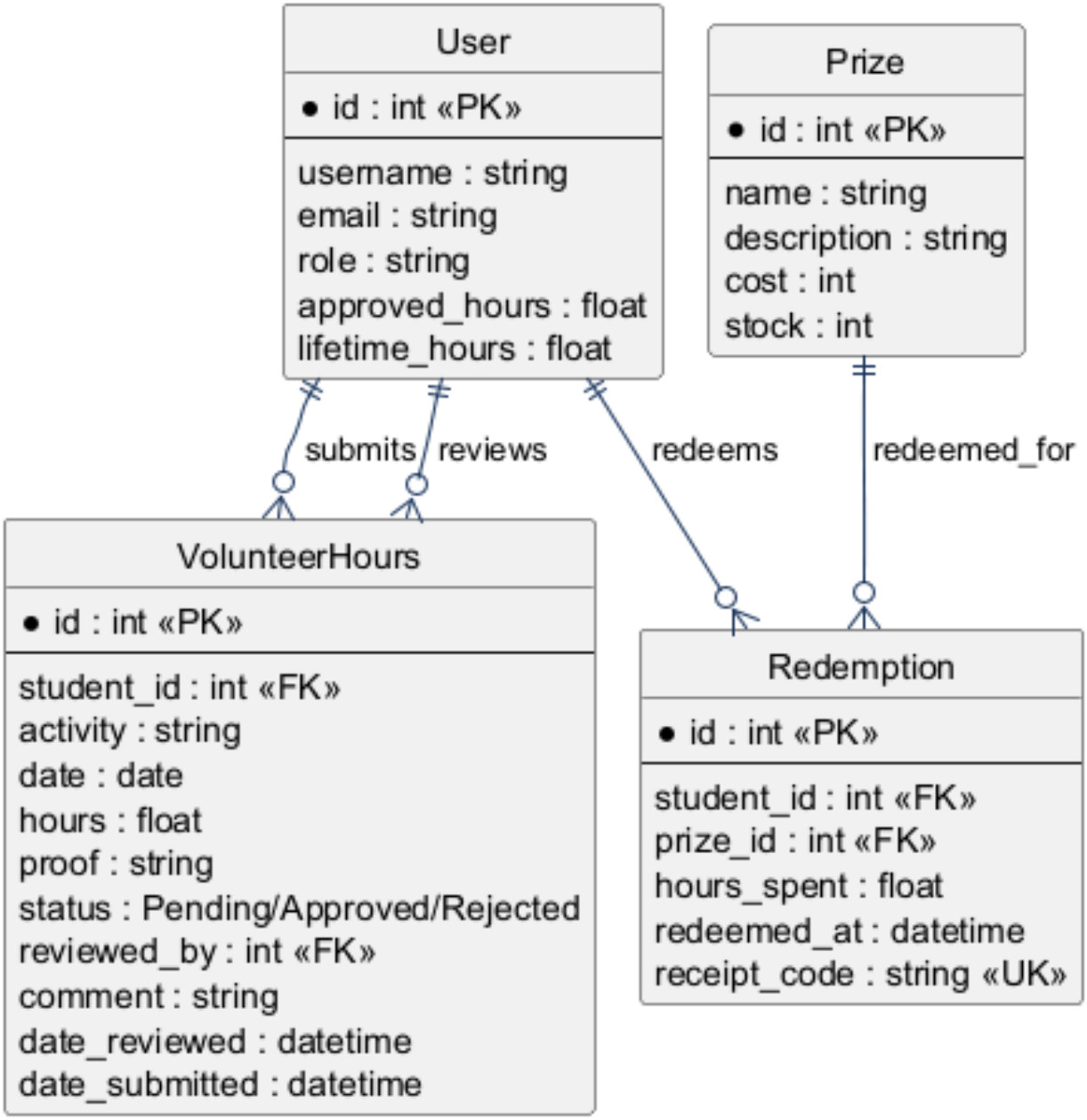
Student Awards is a FastAPI incentive application for tracking student volunteer hours and exchanging approved hours for prizes. Students submit volunteer activities with proof for admin review. Approved submissions increase the student's spendable approved-hours balance and cumulative lifetime-hours total. Students can browse prizes, see stock and point requirements, and redeem an eligible prize with a receipt and QR code. Admins can approve or reject pending submissions; a rejection requires a comment.

Use Case Diagram



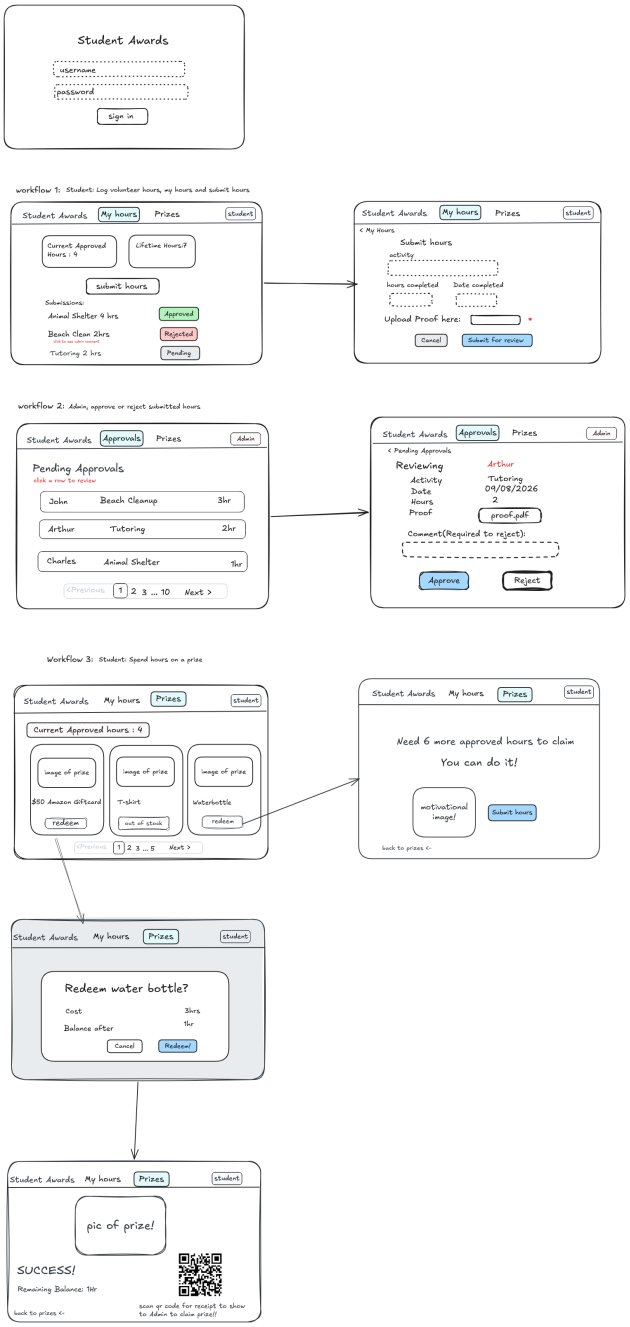
The Student actor signs in, submits hours, views approved hours, browses prizes, and redeems prizes. The Admin actor signs in, views pending approvals, and reviews submissions for approval or rejection. Prize redemption includes browsing the prize list; a points-shortfall message is a conditional extension.

Entity Relationship Diagram



reviewed_by refers to the admin User who reviewed the submission. A Redemption links a student to a prize and stores the hours spent and unique receipt code.

Application Wireframe



Test Plans

The test plans and UAT cases follow the wireframe workflows:

1. Submit volunteer hours as a student.
2. Approve or reject a submission as an admin.
3. Browse prizes and redeem one as a student.

User Acceptance Test Record

The cases below record the reported manual execution against the application wireframe. "Pass (recorded)" reflects the supplied observations.

Test setup: Student account bob; administrator account admin. Before the captured approval, Bob had 2.0 approved hours and 12.0 lifetime hours. The prize-shop scenarios used a 6.0-hour approved balance. The catalog showed the T-shirt out of stock, the $50 giftcard at 10 hours, and the Waterbottle at 3 hours.

ID	Preconditions and test action	Acceptance criteria	Actual result, evidence, and status
UAT-01	Sign in as student bob.	Student session opens with My Hours and Prizes navigation.	Student navigation was visible in supplied screenshots. Pass (recorded).
UAT-02	As bob, submit 4 hours for Animal Shelter with date and proof.	Submission is Pending; balance remains 2.0 before review.	Pending status and unchanged 2.0 balance were shown. Pass (recorded).
UAT-03	As admin, approve Bob's pending 4-hour Animal Shelter submission.	Status becomes Approved; approved/lifetime totals increase by 4.0.	Approved; balances changed 2.0 -> 6.0 and 12.0 -> 16.0. Pass (recorded).
UAT-04	As admin, reject a pending submission with comment blank.	Rejection is blocked and required-comment validation appears.	"A comment is required when rejecting a submission." Pass (recorded).
UAT-05	As admin, reject Bob's 2-hour Planting trees submission with "proof is blurry".	Rejection completes with the supplied comment.	Admin entered the comment and completed rejection. Pass (recorded).
UAT-06	As bob with 6.0 approved hours, view the T-shirt in Prizes.	Balance is 6.0 and T-shirt availability is shown as out of stock.	6.0 balance and "Out of stock" label were shown. Pass (recorded).
UAT-07	As bob, view the $50 giftcard costing 10 hours.	Shortfall of 4.0 hours is stated; submit-hours and return options are available.	Screen stated 4.0 more hours were needed and showed both options. Pass (recorded).

ID	Preconditions and test action	Acceptance criteria	Actual result, evidence, and status
UAT-08	As bob, select the 3-hour Waterbottle and cancel confirmation.	Cancellation returns to Prizes without redeeming.	Confirmation showed 3-hour cost / projected balance; cancel returned to shop without redemption. Pass (recorded).
UAT-09	As bob, confirm the Waterbottle redemption.	Success shows 3.0 remaining hours and receipt/QR.	Success screen showed 3.0 balance, QR, and receipt code. Pass (recorded).

Unit Test Plan

Unit tests isolate service behaviour using a test database and controlled student, admin, and prize records.

ID	Test case	Expected result
UT-01	Submit volunteer hours with a value of zero.	VolunteerHoursService raises a validation error stating hours must be more than zero.
UT-02	Approve a pending 3.5-hour submission.	The submission becomes Approved, and the student's approved and lifetime totals each increase by 3.5 hours.
UT-03	Redeem a 4-hour prize with 10 approved hours and stock of 2.	A redemption records 4 hours spent, the student's balance becomes 6 hours, and prize stock becomes 1.

Integration Test Plan

FastAPI TestClient tests exercise complete HTTP requests against an isolated test database, including authentication, request validation, route responses, and persisted state.

ID	Test case	Expected result
IT-01	Submit volunteer hours through the browser form with a proof upload.	The route redirects with status 303 and stores one Pending submission for the student.
IT-02	Create a volunteer-hours resource through POST /api/hours with form fields and a proof upload.	The route returns 201 with the public resource fields and does not expose the stored proof path.
IT-03	Submit zero hours through the browser form.	Request validation returns 422 and no valid submission is created.
IT-04	Approve a student's pending submission as an admin.	The route redirects with status 303, the submission becomes Approved, and the student's approved balance increases.
IT-05	Redeem a prize as an authenticated student with sufficient hours.	The route redirects with status 303 and prize stock decreases by one.
IT-06	Request the admin page without an authenticated session.	The route returns 401 Unauthorized.
IT-07	Request the admin page as an authenticated student.	The route returns 403 Forbidden.

Run the automated plans from the repository root with:

```
.\.venv\Scripts\python.exe -m pytest tests\test_services.py tests\test_routes.py -q
```

Reference documents supplied for this assignment:

- COMP 3613 Test Plans
- User Acceptance Test Plan reference

Unit Tests

Test file: tests/test_services.py

Command:

```
.\.venv\Scripts\python.exe -m pytest tests\test_services.py -q
```

Result: **3 passed, 1 non-failing deprecation warning.**

Integration Tests

Test file: tests/test_routes.py

Command:

```
.\.venv\Scripts\python.exe -m pytest tests\test_routes.py -q
```

Result: **7 passed, 1 non-failing deprecation warning.**

API Tests (Postman)

The exported collection is student-awards-api.postman_collection.json. It exercises the actual FastAPI routes and cookie-based authentication, including invalid-hours validation, authorization, both form and JSON resource submissions, review, redemption, receipt, and QR responses. The latest Newman run completed with **19 assertions passed and 0 failed** against the freshly seeded disposable database at http://127.0.0.1:8000.

Implementation

The application uses FastAPI and SQLModel with a layered structure: **routes -> services -> repositories -> models**. Routes bind form/API input and delegate business operations to services; repositories handle persistence. Student routes provide hours history, proof upload, prize browsing, shortfall guidance, prize confirmation, redemption, and QR receipts. Admin routes provide a pending-approval list and submission review. The review service enforces Pending-only decisions and requires a comment to reject. Prize redemption checks stock and approved balance before updating the balance and stock.

Repository

https://github.com/aliyahrr/comp3613a1

Test Reports

Unit Test Report

student-awards-unit-report.html

Report generated on 10-Nov-2026 at 16:55:34 by pytest-html v4.2.0

Environment

Python	3.14.5
Platform	Windows-11-10.0.26300-SP0
Packages	• pytest: 9.1.1 • pluggy: 1.6.0
Plugins	• anyio: 4.15.1 • asyncio: 1.4.0 • html: 4.2.0 • metadata: 3.1.1

Summary

3 tests took 276 ms.

(Un)check the boxes to filter the results.

0 Failed, 3 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns 0 Retried, Show all details / Hide all details

Result	Test	Duration	Links
Passed	tests/test_services.py::test_submit_hours_rejects_invalid_hours	65 ms	
Passed	tests/test_services.py::test_review_submission_approves_hours_and_updates_totals	112 ms	
Passed	tests/test_services.py::test_redeem_prize_updates_stock_and_balance	66 ms	

Integration Test Report

student-awards-integration-report.html

Report generated on 10-Nov-2026 at 16:55:35 by pytest-html v4.2.0

Environment

Python	3.14.5
Platform	Windows-11-10.0.26300-SP0
Packages	• pytest: 9.1.1 • pluggy: 1.6.0
Plugins	• anyio: 4.15.1 • asyncio: 1.4.0 • html: 4.2.0 • metadata: 3.1.1

Summary

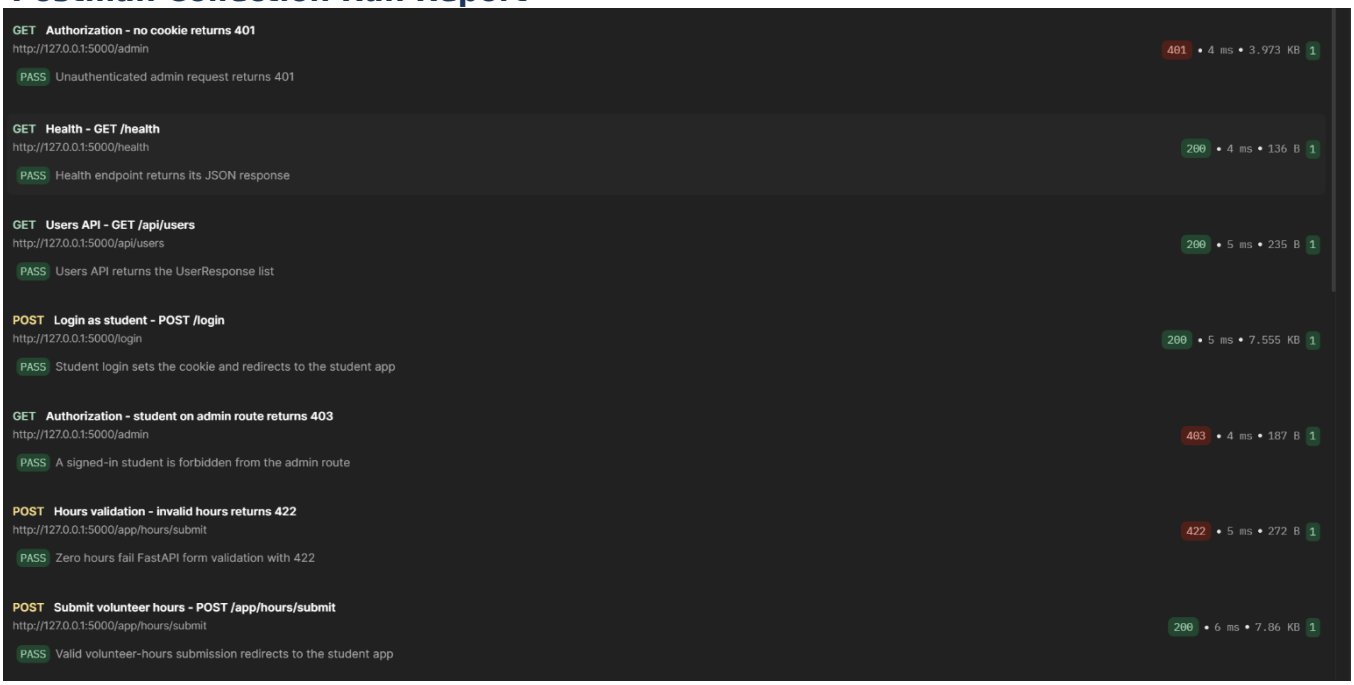
7 tests took 544 ms.

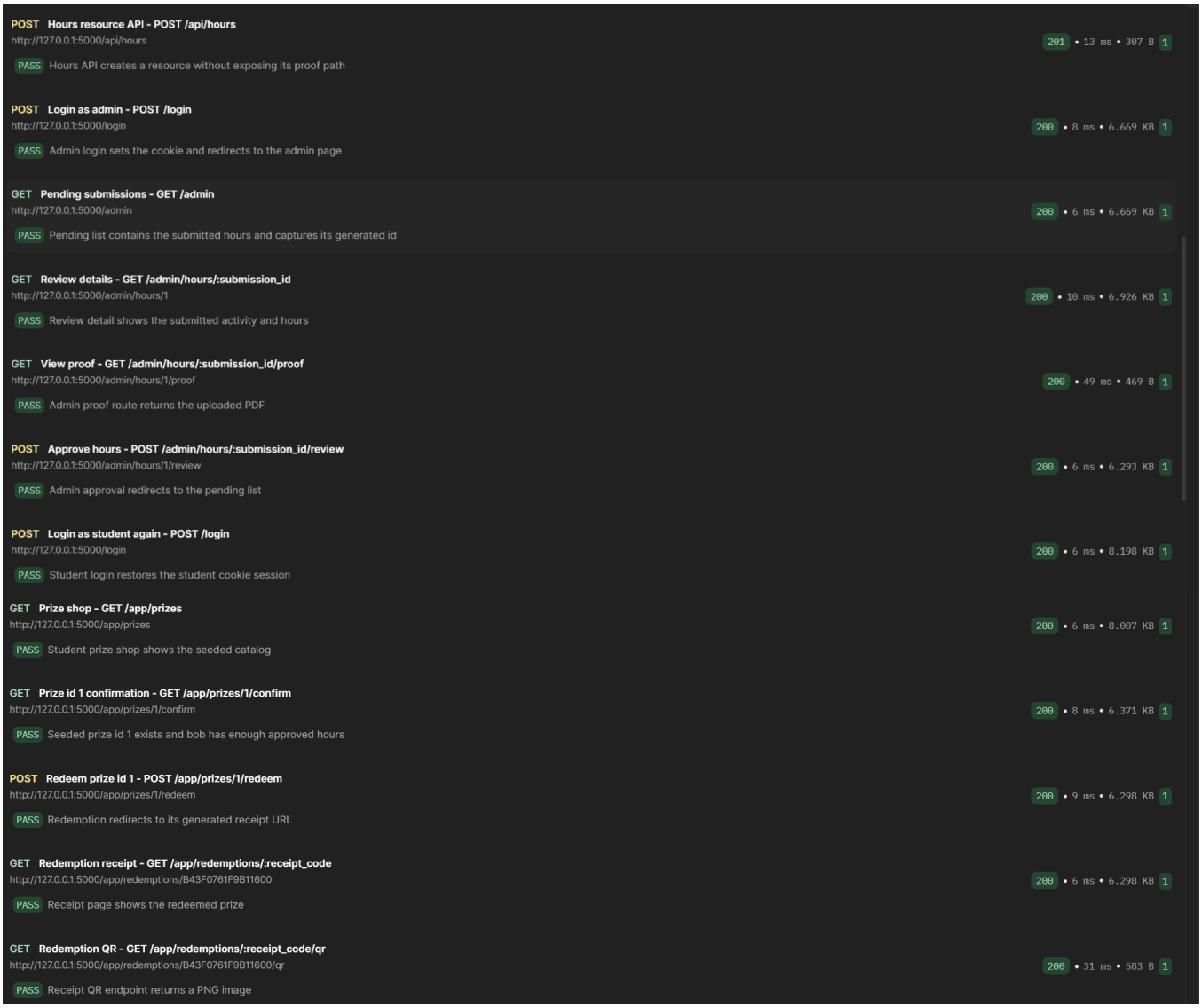
(Un)check the boxes to filter the results.

0 Failed, 7 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns 0 Retried, Show all details / Hide all details

Result	Test	Duration	Links
Passed	tests/test_routes.py::test_student_can_submit_hours	113 ms	
Passed	tests/test_routes.py::test_student_can_create_hours_through_json_resource_route	75 ms	
Passed	tests/test_routes.py::test_zero_hours_request_fails_form_validation	61 ms	
Passed	tests/test_routes.py::test_admin_can_review_a_pending_submission	112 ms	
Passed	tests/test_routes.py::test_student_can_redeem_reward	64 ms	
Passed	tests/test_routes.py::test_unauthenticated_admin_request_returns_401	14 ms	
Passed	tests/test_routes.py::test_student_admin_request_returns_403	62 ms	

Postman Collection Run Report





The Newman report shows 19 requests, 19 test scripts, and 19 assertions with zero failures. The screenshots are arranged in collection order and show requests sent to http://127.0.0.1:5000, including the POST /api/hours resource endpoint.